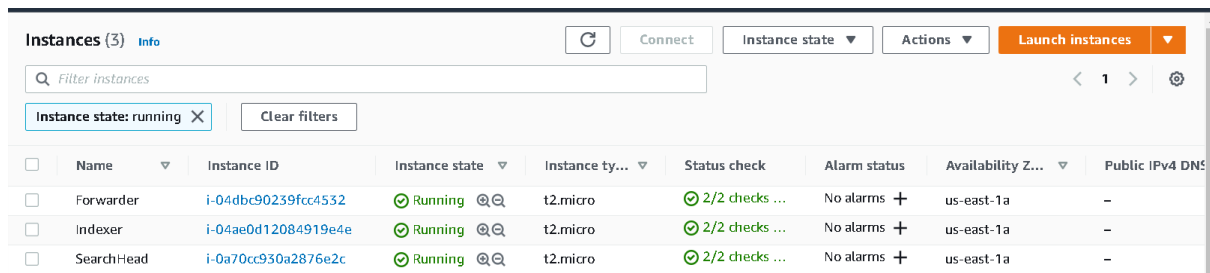


Distributed Architecture without Clustering:

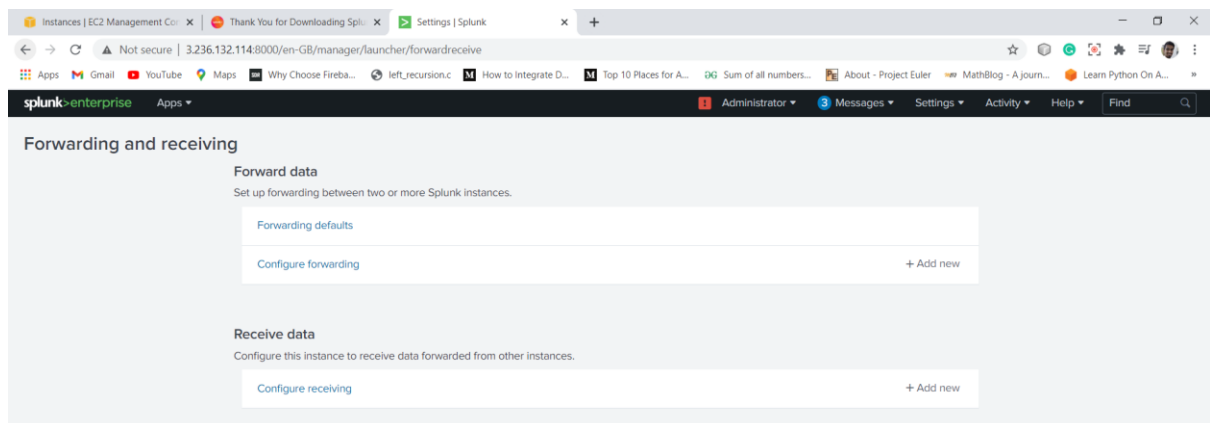
- 1) First create three instances one as Forwarder, second as Indexer and third as Search Head in public subnet for easiness to access. Install and configure Splunk Enterprise in instance you have selected as Search Head and Indexer. Install and configure Splunk Universal Forwarder in the instance you selected as Forwarder.
- 2) In inbound rules of security group of all the three instances allow traffic on port 8000 from source Internet(0.0.0.0/0) or you can be specific also.



The screenshot shows the AWS Management Console 'Instances' page. It displays three instances: 'Forwarder', 'Indexer', and 'SearchHead'. All three are in a 'Running' state. The 'Forwarder' instance has ID 'i-04dbc90239fcc4532', the 'Indexer' has ID 'i-04ae0d12084919e4e', and the 'SearchHead' has ID 'i-0a70cc930a2876e2c'. All are t2.micro instances in the us-east-1a availability zone. The status checks for all instances show '2/2 checks passed'.

	Name	Instance ID	Instance state	Instance ty...	Status check	Alarm status	Availability Z...	Public IPv4 DNS
☐	Forwarder	i-04dbc90239fcc4532	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	Indexer	i-04ae0d12084919e4e	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	SearchHead	i-0a70cc930a2876e2c	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-

- 3) SSH the Indexer open receiving port 9997 from Splunk web page going on *Forwarding and Receiving* and then open the port 9997 in the inbound rules for source Internet in security group of Indexer.



sg-024d48d2ff6907f00 - IndexerSec Actions ▾

Details

Security group name IndexerSec	Security group ID sg-024d48d2ff6907f00	Description Indexer Security	VPC ID vpc-062d320674d5f5241 ↗
Owner 136983541552	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules
Outbound rules
Tags

Inbound rules Edit inbound rules

Type	Protocol	Port range	Source	Description - optional
Custom TCP	TCP	8000	0.0.0.0/0	-
SSH	TCP	22	0.0.0.0/0	-
Custom TCP	TCP	8089	0.0.0.0/0	-
Custom TCP	TCP	9997	0.0.0.0/0	-

4) SSH into the forwarder and give command `./splunk add forward-server index_public_ip : port -auth username:password`.

```
[ec2-user@ip-10-0-1-135 bin]$ sudo ./splunk add forward-server 35.173.1.101:9997 -auth Forwarder:varun@123
Added forwarding to: 35.173.1.101:9997.
```

5) `./splunk add monitor path_of_logs_to_monitor`

```
[ec2-user@ip-10-0-1-135 bin]$ sudo ./splunk add monitor /var/log/messages
Added monitor of '/var/log/messages'.
```

You can see that data is coming from forwarder to Indexer.

The screenshot shows the Splunk Enterprise web interface. A 'Data Summary' window is open, displaying a table of data. The table has columns for 'Host', 'Count', and 'Last Update'. The data row shows 'ip-10-0-1-135' with a count of 937 and a last update time of '13/12/2020 13:51:35.000'.

Host	Count	Last Update
ip-10-0-1-135	937	13/12/2020 13:51:35.000

6) Check if the distributed search is on Search Head. Open the 8089 port of the indexer which will be used as Search peer.

The screenshot shows the 'Distributed search setup' page in the Splunk web interface. The page has a header with the Splunk logo and navigation links. The main content area is titled 'Distributed search setup' and contains a form for configuring distributed search settings. The form includes a section for 'Distributed search set up' with a toggle for 'Turn on distributed search?' set to 'Yes'. Below this is a 'Timeout settings' section with two input fields: 'Status timeout (in seconds)' and 'Server timeout (in seconds)', both set to '10'. At the bottom of the form are 'Cancel' and 'Save' buttons.

Distributed search setup
Distributed search > Distributed search setup

Distributed search set up
Set up distributed search on this page. To view or edit the list of distributed search peers, use the Distributed search peers page in Splunk Settings.

Turn on distributed search? ☒ Yes ☐ No
You must restart your Splunk instance for these settings to take effect.

Timeout settings
Set timeout for distributed search peers.

Status timeout (in seconds)
Set how long to wait for a server to return before removing it from the peers list.

Server timeout (in seconds)
Set how long to wait for a search to return before removing timed out servers from the peers list.

[Cancel](#) [Save](#)

7) Add it indexer as search peer.

The screenshot shows the 'Search peers' page in the Splunk web interface. The page has a header with the Splunk logo and navigation links. The main content area is titled 'Search peers' and contains a table listing the search peers. The table has columns for Peer URI, Splunk instance name, State, Replication status, Cluster label, Health status, Health check failures, Status, and Actions. There is one peer listed with the URI 35.173.110.8089, instance name ip-10-0-1428.ec2.internal, and state Up. A 'New Search Peer' button is located in the top right corner.

Search peers
Distributed search > Search peers

Showing 1 of 1 item

filter

[New Search Peer](#)

Peer URI	Splunk instance name	State	Replication status	Cluster label	Health status	Health check failures	Status	Actions
35.173.110.8089	ip-10-0-1428.ec2.internal	Up	Successful	None	Healthy	None	Enabled Disable	Quarantine Delete

8) Now the forwarder data can be searched from Search Head instance.

The screenshot shows the Splunk Search interface. The search bar contains the query `host=*ip-10-0-1-135*`. The results show 945 events. The table below displays the first five events.

#	Time	Event
>	13/12/2020 13:53:36.000	Dec 13 13:53:36 ip-10-0-1-135 ec2net: [rewrite_aliases] Rewriting aliases of eth0 host = <code>ip-10-0-1-135</code> source = <code>/varlog/messages</code> sourcetype = <code>syslog</code>
>	13/12/2020 13:53:36.000	Dec 13 13:53:36 ip-10-0-1-135 ec2net: [get_meta] Trying to get http://169.254.169.254/latest/meta-data/network/interfaces/macs/02:a8:7c:6b:73:d1/loc al-ipv4s host = <code>ip-10-0-1-135</code> source = <code>/varlog/messages</code> sourcetype = <code>syslog</code>
>	13/12/2020 13:53:36.000	Dec 13 13:53:36 ip-10-0-1-135 ec2net: [get_meta] Getting token for IMDSv2. host = <code>ip-10-0-1-135</code> source = <code>/varlog/messages</code> sourcetype = <code>syslog</code>
>	13/12/2020 13:53:36.000	Dec 13 13:53:36 ip-10-0-1-135 ec2net: [get_meta] Querying IMDS for meta-data/network/interfaces/macs/02:a8:7c:6b:73:d1/local-ipv4s host = <code>ip-10-0-1-135</code> source = <code>/varlog/messages</code> sourcetype = <code>syslog</code>
>	13/12/2020 13:53:36.000	Dec 13 13:53:36 ip-10-0-1-135 dhclient[2719]: bound to 10.0.1.135 -- renewal in 1738 seconds. host = <code>ip-10-0-1-135</code> source = <code>/varlog/messages</code> sourcetype = <code>syslog</code>

Index Clustering Architecture:

9) Create one more Indexer instance and a Master Node using security group used for previous Indexer.

Instances (5) Info								
Filter instances Instance state: running × Clear filters								
☐	Name	Instance ID	Instance state	Instance ty...	Status check	Alarm status	Availability Z...	Public IPv4 DN...
☐	Forwarder	i-04dbc90239fcc4532	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	Indexer	i-04ae0d12084919e4e	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	SearchHead	i-0a70cc930a2876e2c	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	Indexer1	i-051794e2981b42339	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-
☐	Master	i-0827f2f9c58716e54	Running	t2.micro	2/2 checks ...	No alarms +	us-east-1a	-

10) SSH Master node and start Splunk in its web UI go to index clustering in Settings and enable clustering. In configuration select the replication factor, search factor.

Enable Clustering



☒ Master node

The master node coordinates the activities of the peer nodes. It does not store or replicate data (aside from its own internal data).

☐ Peer node

Peer nodes receive and index incoming data. They also replicate data from other nodes in the cluster.

☐ Search head node

The search head manages searches across one or more clusters.

Next

Cancel

Master Node Configuration

Replication Factor

2

The number of copies of raw data that you want the cluster to maintain. A higher replication factor protects against loss of data if peer nodes fail.

Search Factor

1

The number of searchable copies of data the cluster maintains. A higher search factor speeds up the time to recover lost data at the cost of disk space. Must be less than or equal to Replication Factor.

Security Key

Optional

This key authenticates communication between the master and the peers and search heads.

Cluster Label

Optional

Name your cluster using this field. This label is also used to identify this cluster in the Monitoring Console.

Back

Enable Master Node

11) SSH the all the Indexer which are to be made the part of the Index cluster. Enable the 8080 port in the Indexer security group which is going to be used as the replication port. Give the HTTPS URI of the Master to all the Indexers with their management port number.

Enable Clustering



☐ Master node

The master node coordinates the activities of the peer nodes. It does not store or replicate data (aside from its own internal data).

☒ Peer node

Peer nodes receive and index incoming data. They also replicate data from other nodes in the cluster.

☐ Search head node

The search head manages searches across one or more clusters.

Next

Cancel

Peer node configuration



Master URI

E.g. <https://10.152.31.202:8089>

Peer replication port

The port peer nodes use to stream data to each other (Eg: 8080).

Security key

This key authenticates communication between the master and the peers and search heads.

Back

Enable peer node

As soon as all the peers are configured the cluster is working

Clustering: Peer Node

Name ip-10-0-1-128.ec2.internal
Replication port 8080
Master location https://3.235.128.102:8089
Base Generation ID 1

Clustering: Peer Node

Name ip-10-0-1-78.ec2.internal
Replication port 8080
Master location https://3.235.128.102:8089
Base Generation ID 2

Indexer Clustering: Master Node

✓ All Data is Searchable ✓ Search Factor is Met ✓ Replication Factor is Met

2 searchable 0 not searchable Peers 4 searchable 0 not searchable Indexes

Peers (2) Indexes (4) Search Heads (1)

Peer Name	Fully Searchable	Status	Buckets
ip-10-0-1-128.ec2.internal	✓ Yes	Up	17
ip-10-0-1-78.ec2.internal	✓ Yes	Up	11

Search Head Clustering

12) Launch two more instance as Search head and a Deployer using security group used previously for search head. Open the port in inbound rules which is going to be used as replication port.

13) Go to server.conf of the Deployer using SSH client service and do shclustering configuration.

```
[ec2-user@ip-10-0-1-12 bin]$ cd ..
[ec2-user@ip-10-0-1-12 splunk]$ cd etc/system/local
[ec2-user@ip-10-0-1-12 local]$ vi server.conf
```

14) SSH each Search Head Instance go to bin of splunk and execute this command.

```
./splunk init shcluster-config -auth <username>:<password> -mgmt_uri <URI>:<management_port> -
replication_port <replication_port> -replication_factor <n> -conf_deploy_fetch_url
<URL>:<management_port> -secret <security_key>
```

```
ec2-user@ip-10-0-1-44/opt/splunk/bin
[ec2-user@ip-10-0-1-44 bin]$ ./splunk init shcluster-config -auth SearchHead:varun@123 -mgmt_uri https://18.214.39.111:8089 -replication_port 9000 -replication_factor 3 -conf_deploy_fetch_u
rl https://3.239.22.130:8089 -secret varun123
```

15) Make any one Search Head Captain. Using this command

```
./splunk bootstrap shcluster-captain -servers_list
"<URI>:<management_port>,<URI>:<management_port>,..." -auth <username>:<password>
```

16) The cluster is up. See the status of any instance and the members using specific commands.

--see the cluster status

`./splunk show shcluster-status -auth <username>:<password>`

```
[root@ip-10-0-1-33 bin]# ./splunk show shcluster-status -auth searchhead3:varun@123

Captain:
        dynamic_captain : 1
        elected_captain : Sun Dec 13 17:22:29 2020
                id : C1867A0E-0102-4C26-B6DB-37595B840997
        initialized_flag : 1
                label : ip-10-0-1-33.ec2.internal
                mgmt_uri : https://34.235.150.67:8089
        min_peers_joined_flag : 1
        rolling_restart_flag : 0
        service_ready_flag : 1

Members:
        ip-10-0-1-172.ec2.internal
                label : ip-10-0-1-172.ec2.internal
                last_conf_replication : Pending
                mgmt_uri : https://3.237.232.3:8089
                mgmt_uri_alias : https://3.237.232.3:8089
                status : Up
        ip-10-0-1-33.ec2.internal
                label : ip-10-0-1-33.ec2.internal
                mgmt_uri : https://34.235.150.67:8089
                mgmt_uri_alias : https://34.235.150.67:8089
                status : Up
        ip-10-0-1-44.ec2.internal
                label : ip-10-0-1-44.ec2.internal
                last_conf_replication : Pending
                mgmt_uri : https://18.214.39.111:8089
                mgmt_uri_alias : https://18.214.39.111:8089
                status : Up

[root@ip-10-0-1-33 bin]#
```

--see the member status

`./splunk list shcluster-config -auth <username>:<password>`

```
[root@ip-10-0-1-33 bin]# ./splunk list shcluster-config -auth searchhead3:varun@123
config
  adhoc_searchhead:0
  async_replicate_on_proxy:1
  captain_is_adhoc_searchhead:0
  conf_deploy_fetch_url:https://3.239.22.130:8089
  cxn_timeout:60
  decommission_search_jobs_wait_secs:180
  disabled:0
  dispatching_mode:push
  dynamic_captain:1
  heartbeat_period:5
  heartbeat_timeout:60
  id:C1867A0E-0102-4C26-B6DB-37595B840997
  manual_detention:off
  max_peer_rep_load:5
  mode:dynamic_captain
  percent_peers_to_restart:10
  ping_flag:1
  preferred_captain:1
  quiet_period:60
  rcv_timeout:60
  rep_cxn_timeout:60
  rep_max_rcv_timeout:600
  rep_max_send_timeout:600
  rep_rcv_timeout:60
  rep_send_timeout:60
  replication_factor:3
  replication_port:9000
  replication_use_ssl:0
  restart_timeout:600
  rolling_restart:restart
  secret:*****
  send_timeout:60
```